

Task no: 1

date: 29/7/25 Conceptual Design Using ER Model

BANKING MANAGEMENT SYSTEM

Aim: To design an entity Relationship Diagram for a Banking management System.

Steps to draw E-R Diagram:

Step 1: Identifying the main entities.

1. Customer
2. Account
3. Branch
4. Loan
5. Credit Card
6. Banker-info.

Step 2: Defining Attributes for each entity.

1. Customer: Customer-ID (PK), balance, category, Name, address, Phone, email.
2. Account: Account-NO (PK), balance, category
3. Branch: Branch-ID (PK), branch-Name, location, IFSC code, Phone
4. Banker-Info: Banker-Name, Banker-ID (PK), banker-email.
5. Loan: loan-ID (PK), amount, Duration.
6. Credit Card: credit-card number (PK), limit, expiry date.

Step 3: Identifying relation between entities.

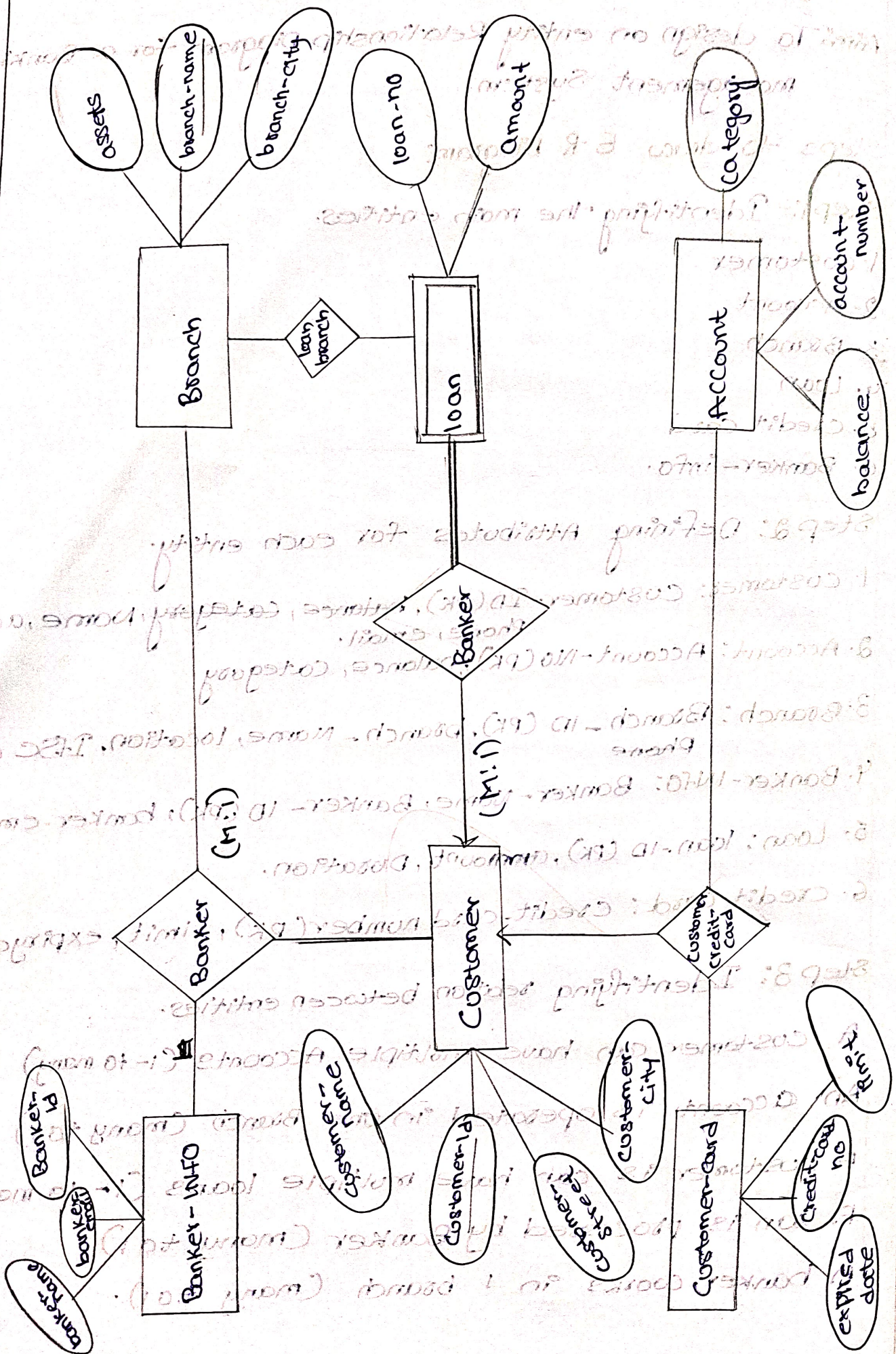
A customer can have multiple Accounts (1-to many)

An account is operated in one Branch (many to 1)

A customer is can have multiple loans (1-to many)

A loan is processed by Banker (many to 1)

A banker works in 1 branch (many to 1).



Step 4: Set Coordinates

using (1:1), (1:N) or (M:N) to indicate how many instances are involved.

Step 5: Draw the ER diagram.

open draw.io website

Draw diagram using:

Rectangle for entities

Ellipse for attributes

Diamonds for Relationships

Lines to connect them

underline the primary keys.

Input:

Banking Management System

Output:

Entity relationship diagram that clearly shows all identified entities with attributes.

All relationships with appropriate coordinates.

foreign keys and keys marked appropriately.

VEL TECH	
EX NO.	1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	15
DATE	

Result:

Hence, the entity-Relationship diagram of Banking management system was successfully drawn using draw.io.

Task NO: 1.2

Date: 28/7/25 ER Diagram to Relational Model

Aim: Convert ER diagram of banking management system into Relational model.

Steps for converting the ER diagram to the table.

- * Entity type becomes a table
- * All single-valued attribute becomes a column for the table.
- * A key attribute of the entity type represented by a separate table primary key.
- * The multivalued attribute is represented by components of separate table.
- * Composite attribute represented by components.
- * Derived attributes are not considered in the table.

VEL TECH	
EX NO.	16
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
DATE	28/7/25

Result: Hence, the conversion of ER diagram of banking management system into Relational model was successfully drawn.

Customer	
	Customer-name
PK FK	Customer-id
	Customer-street
	Customer-city

Banker-INFO	
	banker-name
	banker-email
PK	banker-id

Branch	
	assets
	branch-name
	branch-city

CREDIT - CARD	
	expired-date
PK	Credit-Card-number
	limit

Account	
	Category
	balance
PK PK ₁	Account-number

Loan	
PK	Loan-number
	amount

VEL TECH	
EX NO.	
PERFORMANCE (S)	
RESULT AND ANALYSIS (S)	
VIVA VOCE (S)	
REMARKS (S)	
TOTAL (S)	
DATE	